LAB 07 FUZZ TESTING SE LAB

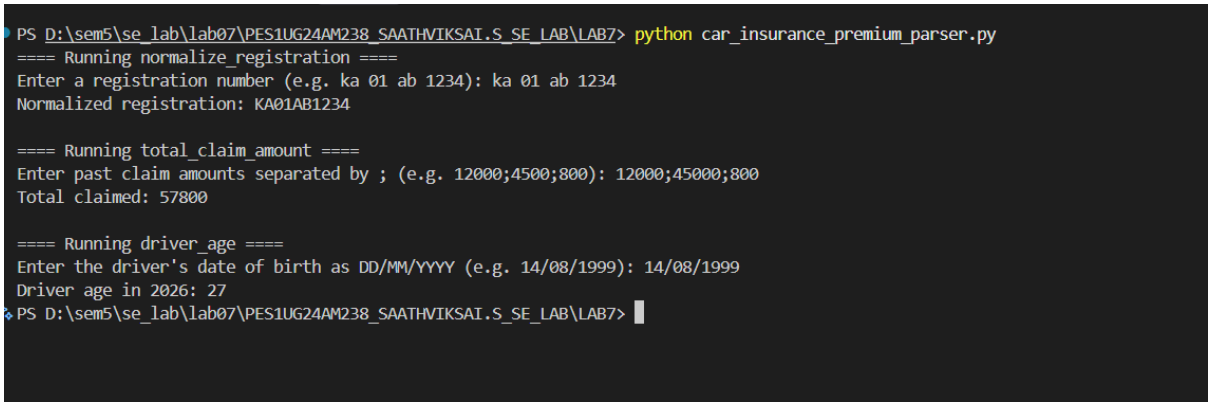
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AIML SEC-D

SCREENSHOTS

1)Run the Buggy Application



2) Failing test cases/errors produced by Hypothesis

```
PS D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7> python -m pytest test_car_insurance_premium_parser.py
============================= test session starts ==============================
platform win32 -- Python 3.13.7, pytest-9.2.0, pluggy-1.7.0
rootdir: D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7
plugins: hypothesis-6.170.15, platformdirs-4.13.6
collected 3 items

test_car_insurance_premium_parser.py FFF                                 [100%]

=================================== FAILURES ===================================
_____________________ test_normalize_registration_no_crash _____________________

s = None

    @given(st.text() | st.none())
    def test_normalize_registration_no_crash(s):
        try:
>           normalize_registration(s)

test_car_insurance_premium_parser.py:7:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

plate = None

    def normalize_registration(plate):
        """
        Removes spaces and dashes from a registration number and upper-cases it.
        Assumes plate is a non-empty string.
        """
>       plate = plate.strip().upper()
                ^^^^^^^^^^
E       AttributeError: 'NoneType' object has no attribute 'strip'

car_insurance_premium_parser.py:6: AttributeError
```

```
During handling of the above exception, another exception occurred:

    @given(st.text() | st.none())
>   def test_normalize_registration_no_crash(s):
                ^^^

test_car_insurance_premium_parser.py:5:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

s = None

    @given(st.text() | st.none())
    def test_normalize_registration_no_crash(s):
        try:
            normalize_registration(s)
        except Exception as e:
            # If the function crashes on unhandled inputs, fail the test and show what caused it
>           assert False, f"normalize_registration crashed with: {e}"
E           AssertionError: normalize_registration crashed with: 'NoneType' object has no attribute 'strip'
E           assert False
E           Failing test case: test_normalize_registration_no_crash(
E               s=None,
E           )

test_car_insurance_premium_parser.py:10: AssertionError
_____________________ test_total_claim_amount_no_crash _____________________
  + Exception Group Traceback (most recent call last):
  |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\test_car_insurance_premium_parser.py", line 13, in test_total_claim_amount_no_crash
  |     def test_total_claim_amount_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\test_car_insurance_premium_parser.py", line 15, in test_total_claim_amount_no_crash
```

```
    |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\test_car_insurance_premium_parser.py", line 22, in test_driver_age_no_crash
    |     driver_age(s)
    |     ^^^^^^^^^^^^^
    |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\car_insurance_premium_parser.py", line 23, in driver_age
    |     return current_year - int(dob.split('/')[2])
    |                           ^^^^^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\test_car_insurance_premium_parser.py", line 24, in test_driver_age_no_crash
    |     assert False, f"driver_age crashed with: {e}"
    | AssertionError: driver_age crashed with: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_driver_age_no_crash(
    |     s=None,
    | )
    +------------------------------------
=========================== short test summary info ============================
FAILED test_car_insurance_premium_parser.py::test_normalize_registration_no_crash - AssertionError: normalize_registration crashed with: 'NoneType' object has no attribute 'strip'
FAILED test_car_insurance_premium_parser.py::test_total_claim_amount_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_car_insurance_premium_parser.py::test_driver_age_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
============================== 3 failed in 2.70s ===============================
PS D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7>
```

```
    |     total_claim_amount(s)
    |     ~~~~~~~~~~~~~~~~~~~^^^
    |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\car_insurance_premium_parser.py", line 16, in total_claim_amount
    |     return sum(int(amount) for amount in text.split(';'))
    |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\car_insurance_premium_parser.py", line 16, in <genexpr>
    |     return sum(int(amount) for amount in text.split(';'))
    |                ~~~^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\test_car_insurance_premium_parser.py", line 17, in test_total_claim_amount_no_crash
    |     assert False, f"total_claim_amount crashed with: {e}"
    | AssertionError: total_claim_amount crashed with: invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_total_claim_amount_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\test_car_insurance_premium_parser.py", line 15, in test_total_claim_amount_no_crash
    |     total_claim_amount(s)
    |     ~~~~~~~~~~~~~~~~~~~^^^
    |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\car_insurance_premium_parser.py", line 16, in total_claim_amount
    |     return sum(int(amount) for amount in text.split(';'))
    |                                           ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\test_car_insurance_premium_parser.py", line 17, in test_total_claim_amount_no_crash
    |     assert False, f"total_claim_amount crashed with: {e}"
    | AssertionError: total_claim_amount crashed with: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_total_claim_amount_no_crash(
    |     s=None,
    | )
    +------------------------------------
_____ test_driver_age_no_crash _____
  + Exception Group Traceback (most recent call last):
  |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\test_car_insurance_premium_parser.py", line 20, in test_driver_age_no_crash
  |     def test_driver_age_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\test_car_insurance_premium_parser.py", line 22, in test_driver_age_no_crash
    |     driver_age(s)
    |     ~~~~~~~~~~^^^
    |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\car_insurance_premium_parser.py", line 23, in driver_age
    |     return current_year - int(dob.split('/')[2])
    |                           ~~~~~~~~~~~~~~~~~^^^
    | IndexError: list index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\test_car_insurance_premium_parser.py", line 24, in test_driver_age_no_crash
    |     assert False, f"driver_age crashed with: {e}"
    | AssertionError: driver_age crashed with: list index out of range
    | assert False
    | Failing test case: test_driver_age_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
```

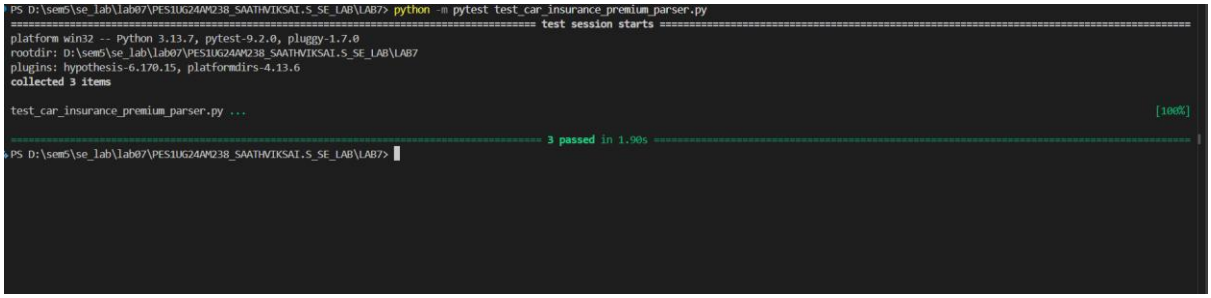
3)After fixing the bugs in car_insurance_premium.py

```
PS D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7> python car_insurance_premium_parser.py
==== Running normalize_registration ====
Enter a registration number (e.g. ka 01 ab 1234): ka 01 ab 2345
Normalized registration: KA 01 AB 2345

==== Running total_claim_amount ====
Enter past claim amounts separated by ; (e.g. 12000;4500;800): 12000;67000;800
Total claimed: 79800

==== Running driver_age ====
Enter the driver's date of birth as DD/MM/YYYY (e.g. 14/08/1999): 14/02/2006
Driver age in 2026: 20
PS D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7>
```

4) After fixing the test file


PS D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7> python -m pytest test_car_insurance_premium_parser.py
test session starts
platform win32 -- Python 3.13.7, pytest-9.2.0, pluggy-1.7.0
rootdir: D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7
plugins: hypothesis-6.170.15, platformdirs-4.13.6
collected 3 items
test_car_insurance_premium_parser.py ...
[100%]
3 passed in 1.90s
PS D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7>